

student_checklist

January 27, 2026 at 09:07 AM

Lecture 6: Workspace Setup Checklist

Follow this checklist to verify you completed each step of the walkthrough.

Part I: Software Installation

- ☐ R version 4.0+ confirmed (`R.version.string`)
- ☐ `remotes` package installed
- ☐ `devtools` package installed
- ☐ `usethis` package installed
- ☐ `testthat` package installed
- ☐ `palmerpenguins` package installed
- ☐ `tidyverse` package installed
- ☐ `rrtools` package installed from GitHub
- ☐ All packages load without errors

Part II: Create Research Compendium

- ☐ Working directory set to a stable location (not Dropbox/OneDrive)
- ☐ Compendium created with `rrtools::use_compendium("penguins.yourname")`
- ☐ New RStudio window opened automatically
- ☐ Analysis structure added with `rrtools::use_analysis()`
- ☐ Folder structure verified with `fs::dir_tree()`

Expected structure:

```
penguins.yourname/
├─ DESCRIPTION
├─ analysis/
├─ data/
├─ raw_data/
├─ derived_data/
├─ figures/
├─ paper/
├─ paper.Rmd
└─ R/
```

Part III: Add Data

- ☐ Palmer penguins data exported to CSV:

```
write.csv(penguins, "analysis/data/raw_data/penguins.csv", row.names = FALSE)
```

- ☐ Data verified: 344 rows, 8 columns

Part IV: Add Analysis Plan Template

- ☐ analysis_plan_template.qmd copied from Lecture 5
- ☐ File saved to analysis/paper/analysis_plan.qmd
- ☐ File exists: file.exists("analysis/paper/analysis_plan.qmd") returns TRUE

Part V: Write Unit Tests

- ☐ testthat initialized with usethis::use_testthat()
- ☐ tests/testthat/ folder exists

Test 1: Data Validation

- ☐ Test file created: usethis::use_test("data-validation")
- ☐ Test code added to tests/testthat/test-data-validation.R:

```
test_that("penguins data file exists and has correct dimensions", {  
  data_path <- "analysis/data/raw_data/penguins.csv"  
  expect_true(file.exists(data_path))  
  penguins_data <- read.csv(data_path)  
  expect_equal(nrow(penguins_data), 344)  
  expect_equal(ncol(penguins_data), 8)  
})
```

Helper Function

- ☐ Function file created: usethis::use_r("clean-data")
- ☐ Function added to R/clean-data.R:

```
#' Remove rows with missing values from penguins data  
#'  
#' @param data A data frame of penguin measurements  
#' @return A data frame with complete cases only  
clean_penguins <- function(data) {  
  data[complete.cases(data), ]  
}
```

Test 2: Function Validation

- ☐ Test file created: usethis::use_test("clean-data")
- ☐ Test code added to tests/testthat/test-clean-data.R:

```
test_that("clean_penguins removes rows with missing values", {
  test_data <- data.frame(
    species = c("Adelie", "Gentoo", "Chinstrap"),
    bill_length = c(39.1, NA, 46.5),
    body_mass = c(3750, 4500, NA)
  )
  cleaned <- clean_penguins(test_data)
  expect_equal(nrow(cleaned), 1)
  expect_equal(cleaned$species, "Adelie")
  expect_false(any(is.na(cleaned)))
})
```

Run Tests

- ☐ Functions loaded: `devtools::load_all()`
- ☐ Tests run: `devtools::test()`
- ☐ Both tests pass (green checkmarks)

Part VI: Version Control and GitHub

- ☐ Git initialized: `usethis::use_git()`
- ☐ Initial commit made
- ☐ GitHub repository created: `usethis::use_github(private = FALSE)`
- ☐ Browser opened to your new repository

Part VII: Final Verification

- ☐ Project structure verified with `fs::dir_tree(recurse = 2)`
- ☐ `devtools::check()` runs with 0 errors

Final Project Structure

Your project should look like this:

```
penguins.yourname/
├─ DESCRIPTION
├─ NAMESPACE
├─ analysis/
│   └─ data/
│       └─ raw_data/
│           └─ penguins.csv
│           └─ derived_data/
│               └─ figures/
│                   └─ paper/
│                       └─ analysis_plan.qmd
│                       └─ paper.Rmd
├─ R/
└─ clean-data.R
```

```
└─ tests/
  └─ testthat.R
    └─ testthat/
      └─ test-clean-data.R
        └─ test-data-validation.R
```

Next Steps

Before January 27:

1. **Fill in your analysis plan** - Edit `analysis/paper/analysis_plan.qmd`
2. **Commit changes regularly** - Use Git pane in RStudio
3. **Push to GitHub** - Keep your repository updated

Troubleshooting

“rrtools not found”

```
remotes::install_github("benmarwick/rrtools")
```

“GitHub PAT not found”

```
usethis::create_github_token()
gitcreds::gitcreds_set()
```

“Test failed”

1. Check file paths are correct
2. Make sure you saved all files
3. Run `devtools::load_all()` before `devtools::test()`

“could not find function ‘clean_penguins’”

```
devtools::load_all()
```

Questions?

- **Office hours:** By appointment
- **Email:** `rgthomas@ucsd.edu` or `maluo@ucsd.edu`
- **Canvas:** Post in discussion board